

PROMPTING STRATEGIES ON ARC: INSIGHTS INTO THE OVER-CORRECTION PARADOX

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ABSTRACT

The Abstraction and Reasoning Corpus (ARC) is a pivotal benchmark for evaluating fluid intelligence in artificial agents. We systematically evaluate various prompting strategies—including Chain-of-Thought (CoT), Self-Consistency (SC), Reflexion, and Program-aided Language models (PAL)—on a subset of ARC tasks ($N = 30$) using `deepseek-chat`. Our results indicate that while structured reasoning significantly outperforms naive induction, performance gains are non-monotonic with respect to prompt complexity. **Self-Consistency** achieved the peak accuracy of **56.67%**, whereas a more complex **Hybrid** framework (integrating SC and Reflexion) resulted in a performance drop to **50.00%**. This degradation reveals an "Over-correction Paradox," where internal self-reflection can override statistically robust consensus. Our study underscores the efficacy of ensemble-based reasoning and identifies the "translation gap" in programmatic synthesis, providing insights into the limitations of purely linguistic metacognition for spatial reasoning.

1 INTRODUCTION

The Abstraction and Reasoning Corpus (ARC), introduced as a benchmark to measure general intelligence, requires agents to perform inductive reasoning from a minimal set of examples. Unlike traditional deep learning tasks that rely on massive datasets, ARC challenges the core ability of systems to acquire new concepts and apply them to unseen spatial grids. While Large Language Models (LLMs) have demonstrated exceptional capabilities in natural language understanding, their performance on zero-shot spatial induction tasks remains a significant hurdle.

Initial experiments using a baseline direct-mapping approach revealed limited success. As shown in our preliminary trials, accuracy fluctuated significantly between 13.33% and 33.33% depending on prompt sensitivity and the presence of system instructions. The model often fails to capture the underlying geometric transformations or suffers from stochastic errors in grid parsing. To bridge this reasoning gap, this paper systematically explores a series of prompt engineering strategies. We begin by implementing Chain-of-Thought (CoT) to decompose complex transformations into logical steps, which improved accuracy to 46.67%.

To further enhance robustness, we introduce Self-Consistency (SC) and Reflexion mechanisms. Our results show that ensemble-based majority voting in Self-Consistency effectively mitigates random failures, achieving a success rate of 56.67%. Within this framework, consensus levels such as "Majority Vote (3/3)" correlate strongly with correctness. Moreover, we implement a multi-stage Reflexion framework to allow the model to self-correct its logic based on internal feedback. We observe that while some tasks are "Passed Directly," others require iterative refinement to overcome initial errors. Finally, we explore a hybrid strategy—combining CoT, SC, and Reflexion—to create a closed-loop reasoning pipeline. Our contributions include:

- **Systematic Evaluation:** A comprehensive assessment of five prompting strategies on the ARC benchmark using the `deepseek-chat` model.

- **Mitigating Model Stubbornness:** The implementation of a “Stern Warning” mechanism within the Reflexion cycle to overcome model “stubbornness” in cases where initial refinements are ineffective.
- **Hybrid Synergy Analysis:** A detailed analysis showing that the hybrid synergy of multi-stage prompts, integrating ensemble consistency and reflexive error correction, significantly outperforms single-stage reasoning.

2 RELATED WORK

The ARC Benchmark. The Abstraction and Reasoning Corpus (ARC) (Chollet, 2019) shifts the focus of AI evaluation from task-specific “skill” to *generalization efficiency*. Solving ARC requires “Core Knowledge” priors—such as objectness and geometry—which are innate to humans but challenging for AI. Recent studies suggest that while Large Language Models (LLMs) possess certain latent reasoning capabilities, they often fail to capture the compositional nature of ARC rules, especially when multiple abstract priors must be applied sequentially (Xu et al., 2023). Our study evaluates how diverse prompting strategies can elicit these latent priors.

Systematic Reasoning via Prompting. LLMs excel at few-shot in-context learning (Brown et al., 2020), inferring patterns without weight updates. *Chain-of-Thought* (CoT) prompting (Wei et al., 2022) further enhances this by eliciting intermediate reasoning steps. However, single-pass CoT is often insufficient for ARC’s high-dimensional grid transformations. To address this, recent research has proposed more structured, search-based reasoning frameworks, such as *Tree of Thoughts* (ToT) (Yao et al., 2023) and *Thought Propagation* (Yu et al., 2023), which treat reasoning as a deliberate path-finding process over multiple hypotheses.

Ensemble Consistency and Iterative Reflection. To improve robustness, *Self-Consistency* (Wang et al., 2022) leverages majority voting across multiple reasoning paths to mitigate stochastic errors. Complementarily, the *Reflexion* framework (Shinn et al., 2023) and the *Self-Refine* approach (Madaan et al., 2023) enable autonomous error correction through linguistic feedback. We integrate these mechanisms to overcome logical “stubbornness” during abstract induction. However, as noted in our experiments, a “refinement plateau” often occurs where models fail to correct initial misinductions despite iterative feedback.

Hybrid and Program-aided Reasoning. The integration of multiple reasoning paradigms—such as combining multi-agent debate or multi-turn reflection with majority voting—has emerged as a robust frontier for complex problem solving (Du et al., 2023). Additionally, translating spatial logic into executable code can bridge the gap between neural perception and symbolic reasoning (Gao et al., 2023). While code-based methods provide deterministic execution for tasks like precise counting, their efficacy in ARC is often bottlenecked by the model’s ability to correctly formalize abstract spatial rules into Python syntax.

3 METHODOLOGY

3.1 PROBLEM FORMULATION

The Abstraction and Reasoning Corpus (ARC) challenges models to induce a generalizable transformation rule $\mathcal{T}$ from a minimal support set of demonstration pairs $\mathcal{D} = \{(X_i, Y_i)\}_{i=1}^n$, where $X_i, Y_i \in \{0, \dots, 9\}^{H \times W}$ are discrete grids. Given a query input X_{test} , the objective is to predict $\hat{Y}_{test} = \mathcal{T}(X_{test})$. We formalize this as a conditional generation task where the model $\mathcal{M}$ aims to maximize the posterior probability:

$$P(\hat{Y}_{test} | \mathcal{D}, X_{test}, \mathcal{P}) \quad (1)$$

where $\mathcal{P}$ denotes the structured prompting strategy. Unlike standard few-shot learning, ARC requires the model to bridge the gap between low-level symbolic representation and high-level geometric abstraction.

3.2 PROMPTING STRATEGIES

We systematically evaluate $\mathcal{M}$ across a hierarchical suite of strategies, ranging from naive heuristics to iterative reasoning frameworks.

3.2.1 BASELINE PROMPTING

To establish a performance floor, we implement two baseline variants. The **Minimalist Baseline** utilizes a direct zero-shot prompt, while the **Naive Context** provides raw training pairs without structural constraints. As we will discuss in Section ??, the performance degradation in the Naive Context variant suggests that unstructured few-shot data increases the *token noise ratio*, distracting the model from global logical invariants.

3.2.2 CHAIN-OF-THOUGHT (CoT) WITH GEOMETRIC PRIORS

To elicit latent spatial reasoning, we design a CoT template that enforces *symbolic grounding* before grid generation. By decomposing the task into explicit steps (see Figure 1), we force the model to verbalize the transformation in terms of objecthood and topology.

Template: CoT Spatial Reasoning

System Instruction:
Act as a world-class ARC solver. Prioritize the following:

- **Objects:** Group connected pixels as distinct entities.
- **Properties:** Identify shapes, colors, and spatial symmetry.
- **Transformations:** Detect movement, rotation, or pattern repetition.

Reasoning Path:

1. *Rule Induction:* Describe the transformation for all training pairs.
2. *Verification:* Ensure the induced rule is consistent across the support set.
3. *Application:* Apply the verified rule to the test input.

Constraint: Encapsulate the final 2D array inside `<result>` tags.

Figure 1: The CoT framework designed to decouple logic discovery from grid reconstruction.

3.2.3 ITERATIVE SELF-REFLECTION (REFLEXION)

To address the limitations of single-pass inference, we implement a *Critic-Refine* loop. We introduce a **Senior Logic Auditor** persona tasked with identifying hidden contradictions through back-testing (see Table 1). If the initial prediction $\hat{Y}_{init}$ is flagged, the model enters a refinement phase to re-evaluate its hypothesis.

Table 1: Iterative Refinement and Audit Instructions

Phase	Instructional Content
Audit	1. Back-test: Apply the proposed rule to Example 1 and verify the output. 2. Geometry: Check if output dimensions align with the transformation logic. 3. Color Logic: Ensure all output colors are justified by the training set.
Refine	1. Fresh Start: Discard the previous hypothesis to avoid confirmation bias. 2. Verification: Mentally re-run the new rule against all training examples. 3. Explanation: Explicitly state what was corrected from the previous attempt.

3.2.4 SELF-CONSISTENCY AND ENSEMBLE VOTING

To mitigate the stochastic nature of autoregressive decoding, we adopt a *Self-Consistency* (SC) framework. We sample $K = 3$ independent reasoning paths and marginalize over the output space to find the majority consensus:

$$\hat{Y}_{final} = \arg \max_Y \sum_{k=1}^K \mathbb{1}[\mathcal{M}(\mathcal{D}, X_{test}, \mathcal{P}) = Y] \quad (2)$$

This strategy aims to filter out high-variance "hallucinated" logic through cross-path verification.

3.2.5 HYBRID STRATEGY: CONSENSUS-THEN-VERIFY (SC + REFLEXION)

To synthesize the strengths of ensemble voting and iterative correction, we propose a *Hybrid* strategy. This approach addresses the residual errors left by individual strategies through a two-stage sequential process:

1. **Consensus Phase:** We generate K independent reasoning paths using the CoT prompt. A majority-vote grid $\hat{Y}_{SC}$ is identified to stabilize the noisy reasoning process.
2. **Verification Phase:** The consensus result $\hat{Y}_{SC}$ is subjected to the *Critic-Refine* pipeline. The Auditor examines $\hat{Y}_{SC}$ as a hypothetical solution. If flaws are found, the model performs a targeted refinement to produce $\hat{Y}_{hybrid}$.

This strategy effectively treats the ensemble result as a "informed prior" that the reflection stage then fine-tunes (see Figure 2).

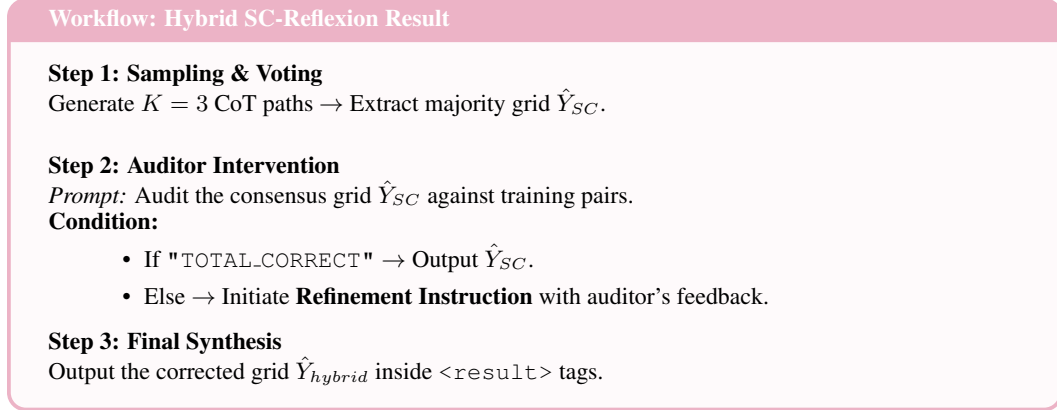


Figure 2: The Hybrid framework (SC+Reflexion). It achieves the robustness of collective intelligence while maintaining the rigor of formal verification.

3.3 PROGRAMMATIC SYNTHESIS: THE TRANSLATION GAP

As a deterministic alternative, we evaluate *Program-aided Language Models* (PAL). Instead of natural language, the model synthesizes a Python function (see Figure 3). This approach tests whether the model can formalize spatial intuition into algorithmic syntax.

4 EXPERIMENTS AND RESULTS

4.1 EXPERIMENTAL SETUP

We evaluate the proposed prompting strategies on a representative subset of the ARC evaluation set consisting of $N = 30$ tasks. All experiments are conducted using the `deepseek-chat` model. For deterministic strategies such as the Minimalist Baseline and CoT, we utilize a low temperature

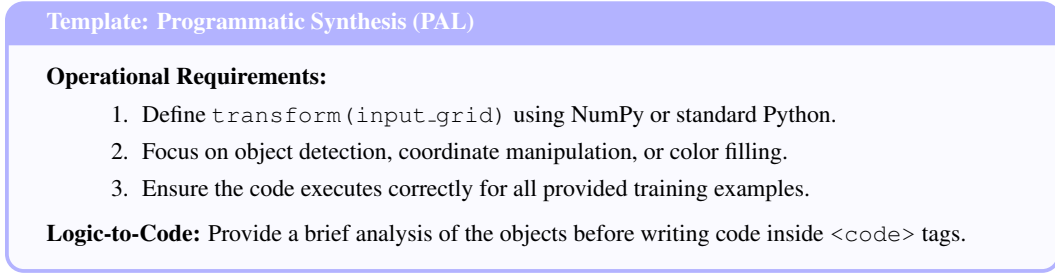


Figure 3: The PAL framework designed to leverage deterministic execution engines.

($T = 0.1$) to ensure stable output. For ensemble and iterative methods including Self-Consistency (SC) and Hybrid, we set $T = 1.0$ and $K = 3$ sampling paths to encourage diversity in the model’s reasoning space.

4.2 MAIN RESULTS AND PERFORMANCE TREND

The empirical results, summarized in Table 2, reveal a non-linear relationship between prompt complexity and task accuracy.

Table 2: Accuracy of different prompting strategies on ARC ($N = 30$).

Strategy	Correct Tasks	Accuracy (%)	Process Status
Minimalist Baseline	10	33.33%	Direct Prompt
Naive Context	4	13.33%	Raw Few-shot
Chain-of-Thought (CoT)	14	46.67%	Two-stage Reasoning
Reflexion	16	53.33%	Single-path Refine
Self-Consistency (K=3)	17	56.67%	Majority Vote
Hybrid (CoT+SC+Reflexion)	15	50.00%	Consensus + Refine
Programmatic (PAL)	10	33.33%	Code Synthesis

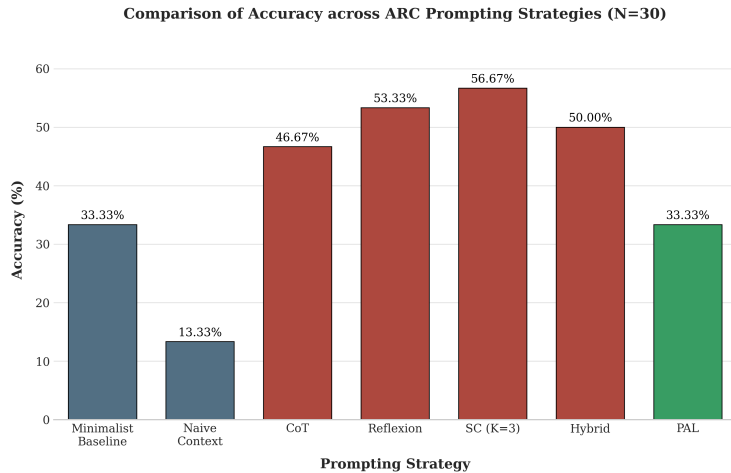


Figure 4: Accuracy trend across evaluated strategies. The performance peaks at Self-Consistency, while the Hybrid approach suffers a degradation due to over-correction.

4.3 IN-DEPTH ANALYSIS AND DISCUSSION

4.3.1 THE PARADOX OF HYBRID REFINEMENT

A significant observation is that the *Hybrid* strategy, which combines the robustness of Self-Consistency with the rigor of Reflexion, yielded a lower accuracy (50.00%) than pure SC (56.67%). Our analysis of the process logs suggests that the model often encounters an **Over-correction Plateau**. In multiple instances where the SC phase reached a correct majority consensus (e.g., Task 1, 3, 7), the subsequent Auditor intervention incorrectly flagged the result as flawed. Given the "Stern Warning" in the refinement prompt, the model frequently abandoned its correct spatial logic in favor of erroneous adjustments.

4.3.2 SYMBOLIC GROUNDING VS. NAIVE INDUCTION

The contrast between the **Minimalist Baseline** (33.33%) and **Naive Context** (13.33%) highlights the sensitivity of LLMs to ARC grid representations. Providing raw training pairs without structural guidance leads to a higher *token noise ratio*, distracting the model from identifying the core geometric transformation. Conversely, the introduction of **CoT** (46.67%) demonstrates that forcing the model to verbalize rules before predicting the grid significantly mitigates these grounding errors].

4.3.3 THE TRANSLATION GAP IN PROGRAMMATIC SYNTHESIS

The **PAL** approach (33.33%) struggled with "Code Refinement Failure" in the majority of tasks (e.g., Tasks 1, 5, 9, 20-29). While the model could often identify the transformation rule in natural language, it failed to formalize these concepts into valid Python logic or NumPy-style indexing, revealing a persistent gap between spatial perception and syntactic implementation.

4.4 CONCLUSION OF RESULTS

In summary, while iterative reflection and code synthesis provide deterministic rigor, they are prone to error propagation in the absence of external execution feedback. The **Self-Consistency** mechanism remains the most robust strategy for ARC, as it leverages the "wisdom of the crowd" in the model's latent space to filter out high-variance reasoning failures.

5 DISCUSSION

5.1 THE FRAGILITY OF SELF-CORRECTION WITHOUT VERIFICATION

Our results reveal a critical counter-intuitive phenomenon: the **Hybrid** strategy (50.00%) underperforms the simpler **Self-Consistency** (56.67%). This degradation highlights the *Fragility of Self-Correction* in LLMs for non-linguistic tasks.

- **Instructional Dominance:** When the Auditor persona issues a critique, the model's "compliance bias" often outweighs its internal logical consistency.
- **Absence of Ground Truth:** Unlike mathematical or coding tasks where a compiler or solver can provide objective feedback, ARC refinement relies purely on latent reasoning, which is prone to recursive error propagation once a correct consensus is discarded].

5.2 THE SPATIAL-SEMANTIC TRANSLATION GAP

The stagnation of the **PAL** approach (33.33%) at the baseline level provides empirical evidence of a "Translation Gap". While the model correctly verbalizes spatial transformations in natural language (e.g., in the CoT stage), it fails to map these abstract concepts—such as "connectivity" or "containment"—into precise NumPy-style indexing or iterative loops. This suggests that for ARC, the bottleneck is not the reasoning itself, but the formalization of spatial intuition into symbolic code.

5.3 NOISE SENSITIVITY IN FEW-SHOT LEARNING

The significant performance drop in the **Naive Context** variant (13.33%) suggests that for high-entropy symbolic tasks like ARC, raw data samples act as high-frequency noise. Without the "symbolic grounding" provided by CoT or structural delimiters, the model fails to isolate the invariant transformation rule from the distracting pixel-level variations.

6 CONCLUSION

In this study, we systematically explored a hierarchy of prompting strategies for the Abstraction and Reasoning Corpus (ARC). Our experiments demonstrate that **Self-Consistency** (56.67%) currently provides the most robust mechanism for symbolic induction by leveraging the "wisdom of the crowd" in the model's latent space.

Crucially, we identified the **Hybrid Paradox**: adding a reflection layer over a consensus-based result can lead to performance degradation due to over-correction. These findings suggest that future efforts in solving ARC should prioritize the development of **execution-guided refinement**—where models receive objective feedback from external environments—rather than relying solely on internal self-reflection. Bridging the gap between spatial intuition and programmatic formalization remains the most promising frontier for achieving human-level general intelligence in symbolic reasoning.

REFERENCES

- Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. *Advances in neural information processing systems*, 33:1877–1901, 2020.
- François Chollet. On the measure of intelligence. *arXiv preprint arXiv:1911.01547*, 2019.
- Yilun Du, Shuang Li, Antonio Torralba, Joshua B Tenenbaum, and Igor Mordatch. Improving factuality and reasoning in language models through multiagent debate. In *Forty-first International Conference on Machine Learning*, 2023.
- Luyu Gao, Aman Madaan, Shuyan Zhou, Uri Alon, Pengfei Liu, Yiming Yang, Jamie Callan, and Graham Neubig. Pal: Program-aided language models. In *International Conference on Machine Learning*, pp. 10764–10799. PMLR, 2023.
- Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegreffe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, et al. Self-refine: Iterative refinement with self-feedback. *Advances in Neural Information Processing Systems*, 36:46534–46594, 2023.
- Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. *Advances in Neural Information Processing Systems*, 36:8634–8652, 2023.
- Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. *arXiv preprint arXiv:2203.11171*, 2022.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, et al. Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems*, 35:24824–24837, 2022.
- Yudong Xu, Wenhao Li, Pashootan Vaezipoor, Scott Sanner, and Elias B Khalil. Llms and the abstraction and reasoning corpus: Successes, failures, and the importance of object-based representations. *arXiv preprint arXiv:2305.18354*, 2023.
- Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Tom Griffiths, Yuan Cao, and Karthik Narasimhan. Tree of thoughts: Deliberate problem solving with large language models. *Advances in neural information processing systems*, 36:11809–11822, 2023.

Junchi Yu, Ran He, and Rex Ying. Thought propagation: An analogical approach to complex reasoning with large language models. *arXiv preprint arXiv:2310.03965*, 2023.